

Answer Key & Teacher Notes

Session 1 – Dividing by a Unit Fraction • Session 2 – Dividing a Fraction by a Whole Number

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Dividing by a Unit Fraction

TALK PROMPTS

- How can dividing by $\frac{1}{4}$ give an answer greater than the number you started with?
- The yarn is only 15 meters long, but $15 \div \frac{1}{5} = 75$. Why does that make sense?

PARTNER TASK

Tape and Yarn. One partner writes $3 \div \frac{1}{10}$ for 3 yards of tape cut into $\frac{1}{10}$ -yard pieces; the other rewrites it as multiplication and solves (30). Switch roles with Craft Table: 10 meters of yarn, $\frac{1}{5}$ meter per bracelet (50).

WATCH FOR

- Flipping the dividend instead of the divisor ($4 \div \frac{1}{3}$ written as $\frac{1}{4} \times \frac{1}{3} = \frac{1}{12}$).
- Dividing by the denominator ($8 \div \frac{1}{2} = 4$): finding half of 8 instead of how many halves are in 8.
- Expecting every quotient to be smaller than the dividend.

SESSION 2 • TEACHER NOTES

Dividing a Fraction by a Whole Number

TALK PROMPTS

- Why does sharing a fraction among more people make every share smaller?
- The trail mix was already less than one pound. Why is each hiker's share still smaller than $\frac{3}{4}$ pound?

PARTNER TASK

Lab Day. One partner finds $\frac{7}{8} \div 5$ ($\frac{7}{40}$ liter per beaker); the other checks by multiplying the share by 5. Switch roles with Fabric Strip: $\frac{6}{7}$ meter cut into 3 pieces ($\frac{2}{7}$ meter).

WATCH FOR

- Multiplying by the whole number instead of its reciprocal ($\frac{3}{4} \div 10$ written as $\frac{3}{4} \times 10 = \frac{30}{4}$).
- Flipping the fraction instead of the whole-number divisor ($\frac{3}{4} \div 3$ written as $\frac{4}{3} \times 3$).
- Sharing the whole amount before subtracting what was poured out.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.NS.A.1	1	2	1	1	2	2	3	2	3	1	1
S1 • Form B	6.NS.A.1	1	2	1	1	2	2	3	2	3	1	1
S2 • Form A	6.NS.A.1	1	2	1	1	2	2	3	2	3	1	1
S2 • Form B	6.NS.A.1	1	2	1	1	2	2	3	2	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

unit fraction: A fraction with **1** as its numerator.

reciprocal: The number you multiply by to get a product of **1**.

quotient: The **answer** to a division problem.

- Dividing by a unit fraction asks how many **pieces** of that size fit into the number.
- Dividing by $1/4$ is the same as multiplying by **4**, so $3 \div 1/4 = 3 \times 4 = 12$.
- The quotient is **greater** than the whole number you started with.

Example: Each pound holds **4** fourths. 3 pounds hold $3 \times 4 = 12$ fourths. Check: $12 \times 1/4 = 3$ pounds. Answer: $3 \div 1/4 = 12$ balls of clay

YOU TRY

- a. 12 b. 25
- $2 \div 1/8 = 2 \times 8 = 16$ pieces

EXIT TICKET

- a. 24 b. 18
- $5 \div 1/3 = 5 \times 3 = 15$ bracelets
- A** (7×5)

PRACTICE FORM A

- a. 16 b. 12 c. 30 d. 15 e. 40 f. 42
- a. $8 \div 1/4 = 8 \times 4 = 32$ projects b. $2 \div 1/8 = 2 \times 8 = 16$ cubes
- $12 \div 1/3 = 12 \times 3 = 36$ beds
- $6 \div 1/8 = 6 \times 8 = 48$ cups
- No. $8 \div 1/2 = 8 \times 2 = 16$ bottles. Mia found half of 8 ($8 \div 2$) instead of how many halves are in 8.
- a. $2 \div 1/12 = 24$ pieces b. $24 - 8 = 16$ pieces
- a. $15 \div 1/5 = 75$ bracelets b. $15 \div 1/3 = 45$ bracelets
c. Each bracelet uses a bigger piece, so fewer pieces fit in the same 15 meters.
- a. $9 \times 3 = 27$ b. Each whole holds 3 thirds, so 9 wholes hold $9 \times 3 = 27$ thirds. We count small pieces, so $27 > 9$.
- Rosa flipped the dividend 4 instead of the divisor $1/3$. Keep 4, change $\div$ to $\times$, flip $1/3$ to 3: $4 \times 3 = 12$. There are 12 thirds in 4 wholes.
- B** (20)
- C** (greater than 6)

Self-check word DIVIDE

PRACTICE FORM B

- a. 14 b. 8 c. 24 d. 20 e. 60 f. 40
- a. $6 \div 1/3 = 6 \times 3 = 18$ feeders b. $3 \div 1/4 = 3 \times 4 = 12$ packets
- $10 \div 1/4 = 10 \times 4 = 40$ beds
- $4 \div 1/6 = 4 \times 6 = 24$ cups
- No. $9 \div 1/2 = 9 \times 2 = 18$ bottles. Theo found half of 9 ($9 \div 2$) instead of how many halves are in 9.
- a. $3 \div 1/10 = 30$ pieces b. $30 - 12 = 18$ pieces
- a. $10 \div 1/5 = 50$ bracelets b. $10 \div 1/2 = 20$ bracelets
c. Each bracelet uses a bigger piece, so fewer pieces fit in the same 10 meters.
- a. $7 \times 5 = 35$ b. Each whole holds 5 fifths, so 7 wholes hold $7 \times 5 = 35$ fifths. We count small pieces, so $35 > 7$.
- Amir flipped the dividend 4 instead of the divisor $1/5$. Keep 4, change $\div$ to $\times$, flip $1/5$ to 5: $4 \times 5 = 20$. There are 20 fifths in 4 wholes.
- D** (18)
- B** (9)

Self-check word GROUPS

GUIDED NOTES

reciprocal: Two numbers are reciprocals when their product is **1**.

multiplicative inverse: Another name for the **reciprocal**.

divisor: The number you **divide** by.

- Dividing a fraction by a whole number **shares** it into equal parts.
- To divide by 4, multiply by its reciprocal, **$\frac{1}{4}$** .
- Each share is **less** than the fraction you started with.

Example: Write 4 as a fraction: $2/5 \div 4/1$. Multiply by the reciprocal: $2/5 \times 1/4 = 2/20$. Simplify: $2/20 = 1/10$.

Answer: Each person gets **$1/10$** pound of rice.

YOU TRY

- a. $1/12$ b. $2/7$
- $5/8 \div 5 = 5/8 \times 1/5 = 5/40 = 1/8$ of the pizza

EXIT TICKET

- a. $1/10$ b. $3/7$
- $3/4 \div 6 = 3/4 \times 1/6 = 3/24 = 1/8$ L
- B** ($2/9 \times 1/4$)

PRACTICE FORM A

- a. $1/6$ b. $1/5$ c. $2/5$ d. $1/10$ e. $4/9$ f. $1/12$
- a. $3/4 \div 4 = 3/4 \times 1/4 = 3/16$ pound b. $2/3 \div 10 = 2/3 \times 1/10 = 2/30 = 1/15$ gallon
- $3/5 \div 10 = 3/5 \times 1/10 = 3/50$ kg
- $6/7 \div 3 = 6/7 \times 1/3 = 6/21 = 2/7$ L
- No. $3/4 \div 4 = 3/4 \times 1/4 = 3/16$ yard. Priya multiplied by 4. Each piece must be shorter than the whole $3/4$ -yard string.
- a. $7/12 - 3/12 = 4/12 = 1/3$ L b. $1/3 \div 4 = 1/12$ L
- a. $7/8 \div 7 = 1/8$ kg b. $7/8 \div 14 = 1/16$ kg c. Doubling the number of trays halves each share, because the same dough is split twice as many ways.
- $8/9 \div 4 = 8/9 \times 1/4 = 8/36 = 2/9$ m. Check: $2/9 \times 4 = 8/9$.
- Caleb multiplied by 10 instead of by its reciprocal $1/10$. $3/4 \div 10 = 3/4 \times 1/10 = 3/40$. The share must be less than $3/4$, and $30/4$ is greater.
- B** ($1/8$)
- A** ($1/8$)

Self-check word SHARING

PRACTICE FORM B

- a. $1/8$ b. $1/6$ c. $1/4$ d. $2/9$ e. $1/7$ f. $1/18$
- a. $5/6 \div 5 = 5/6 \times 1/5 = 5/30 = 1/6$ pound b. $3/4 \div 8 = 3/4 \times 1/8 = 3/32$ gallon
- $2/5 \div 8 = 2/5 \times 1/8 = 2/40 = 1/20$ kg
- $9/10 \div 3 = 9/10 \times 1/3 = 9/30 = 3/10$ L
- No. $2/3 \div 4 = 2/3 \times 1/4 = 2/12 = 1/6$ yard. Malik multiplied by 4. Each piece must be shorter than the whole $2/3$ -yard string.
- a. $9/10 - 3/10 = 6/10 = 3/5$ L b. $3/5 \div 3 = 1/5$ L
- a. $7/8 \div 5 = 7/40$ L b. $7/8 \div 10 = 7/80$ L c. Doubling the number of beakers halves each share, because the same solution is split twice as many ways.
- $6/7 \div 3 = 6/7 \times 1/3 = 6/21 = 2/7$ m. Check: $2/7 \times 3 = 6/7$.
- Jordan multiplied by 2 instead of by its reciprocal $1/2$. $4/9 \div 2 = 4/9 \times 1/2 = 4/18 = 2/9$. The share must be less than $4/9$, and $8/9$ is greater.
- B** ($1/5$)
- C** (smaller than the fraction)

Self-check word PORTION

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 · DIVIDING BY A UNIT FRACTION

1. 12 (each whole has 3 thirds; $4 \times 3 = 12$)
2. a. 12 b. 15 c. 32 d. 18 e. 21 f. 60 g. 35 h. 36
3. $6 \div \frac{1}{4} = 6 \times 4 = 24$ bags
4. $5 \div \frac{1}{6} = 30$ pieces; $30 - 18 = 12$ pieces left
5. She found half of 8 ($8 \div 2$), not how many halves are in 8. Each whole has 2 halves, so $8 \div \frac{1}{2} = 8 \times 2 = 16$.

SHEET 2 · DIVIDING A FRACTION BY A WHOLE NUMBER

1. a. $\frac{1}{6}$ b. $\frac{2}{5}$ c. $\frac{1}{8}$ d. $\frac{1}{9}$ e. $\frac{3}{7}$ f. $\frac{1}{12}$
2. a. $\frac{1}{2} \div 4 = \frac{1}{8}$ lb b. $\frac{5}{6} \div 5 = \frac{1}{6}$ L
3. $\frac{3}{5} \div 6 = \frac{3}{5} \times \frac{1}{6} = \frac{3}{30} = \frac{1}{10}$ gallon
4. $\frac{2}{3} \div 4 = \frac{2}{3} \times \frac{1}{4} = \frac{2}{12} = \frac{1}{6}$ pound
5. less than $\frac{5}{8}$ ($\frac{5}{8} \div 4 = \frac{5}{32}$; sharing into 4 parts makes each part smaller)
6. **B** ($\frac{5}{6} \times \frac{1}{5}$)
7. **A** ($\frac{1}{9}$)